US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12398870
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Shu; Jinbo

Projection device

Abstract

A projection device includes at least two projection lamps and a base. The base includes at least two connecting portions. The at least two connecting portions extend in different directions. At least one of the at least two projection lamps is disposed on each of the connecting portions. The base is allowed to mount the at least two projection lamps, thereby improving mounting efficiency and space utilization.

Inventors: **Shu; Jinbo (Zhejiang, CN)**

Applicant: **Shu; Jinbo** (Zhejiang, CN)

Family ID: **1000008657530**

Appl. No.: **19/003687**

Filed: **December 27, 2024**

Foreign Application Priority Data

CN	202422991728.7	Dec. 04, 2024

Publication Classification

Int. Cl.: **F21V21/116** (20060101); **F21S8/08** (20060101); **F21V21/08** (20060101); **G03B21/14** (20060101)

U.S. Cl.:

CPC **F21V21/116** (20130101); **F21S8/081** (20130101); **G03B21/145** (20130101); F21V21/0824 (20130101)

Field of Classification Search

CPC: G03B (21/145); F21V (21/0824)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5655829	12/1996	Lin	362/431	F21V 21/30
10042238	12/2017	Zhang	N/A	G03B 21/003
10145520	12/2017	Tang	N/A	F21V 21/0824
10146120	12/2017	Zhang	N/A	G03B 21/001
D1005550	12/2022	Zhou	D26/63	N/A
D1032066	12/2023	Wang	D26/106	N/A
D1052773	12/2023	Wang	D26/63	N/A
2014/0211459	12/2013	Toner	362/396	F21V 33/00
2016/0026073	12/2015	Zhang	353/101	G03B 21/2033
2017/0082254	12/2016	Zhang	N/A	A63H 27/10
2017/0219176	12/2016	Chang	N/A	F21V 5/04
2017/0227200	12/2016	Zhang	N/A	F21V 21/30
2018/0283663	12/2017	Zhang	N/A	F21V 14/06

Primary Examiner: Garlen; Alexander K

Attorney, Agent or Firm: HOWARD M COHN and Associates, LLC

Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to a technical field of projection devices, and in particular to a projection device.
BACKGROUND
(2) Projection devices adopt an optical projection principle and high-brightness light sources to project image structures (pattern or text content) on a projection element thereof onto a wall, floor or screen.
(3) However, only one projection lamp is disposed on a base of a projection device in the related art. When an effect of multiple projection patterns is required, additional mounting bases are required to mount additional projection lamps, and each of the mounting bases occupies a certain mounting space. When mounting the projection lamps, mounting efficiency is reduced and a space utilization rate is low.
SUMMARY
(4) The embodiments of the present disclosure provide a projection device, which realizes mounting of projection lamps on a base, thereby improving mounting efficiency and space utilization.
(5) The projection device of the present disclosure comprises at least two projection lamps and a base.
(6) The base comprises at least two connecting portions. The at least two connecting portions extend in different directions. At least one of the at least two projection lamps is disposed on each of the connecting portions.

(7) In some embodiments, the base comprises a connecting piece and a fixing piece. The connecting piece comprises the at least two connecting portions and a mounting portion. The at least two connecting portions are connected to the mounting portion. The fixing piece is connected to one end of the mounting portion away from the at least two projection lamps. The fixing piece fixes the at least two projection lamps to a mounting surface or an external device.
(8) In some embodiments, the at least two projection lamps are respectively rotatably connected to the at least two connecting portions.
(9) In some embodiments, the at least two connecting portions comprise connection positions respectively connected to the at least two projection lamps. In a front-rear direction of the projection device, the connection positions are located on a same plane.
(10) In some embodiments, the connecting piece further comprises spherical portions and mounting pieces, each of the spherical portions are connected to one end of a corresponding one of the at least two connecting portions. Each of the projection lamps comprises a housing, each housing define a mounting groove, and a groove wall of each mounting groove is spherical.
(11) Each of the mounting pieces defines a through hole. An aperture of each through hole is less than an outer diameter of each of the spherical portions. Each of the mounting pieces is detachably connected to each housing to rotatably connect each of the spherical portions in each mounting groove. Each of the connecting portions extends outward from each through hole.
(12) In some embodiments, the connecting piece comprises a main body portion and hemispherical coves. The main body portion comprises the mounting portion, the at least two connecting portions, and hemispherical portions respectively connected to the at least two connecting portions. The hemispherical covers are respectively detachably connected to the hemispherical portions to form the spherical portions.
(13) In some embodiments, the hemispherical covers are respectively connected to the hemispherical portions through screws.
(14) In some embodiments, each housing comprises a housing body and arc-shaped connecting plates. The arc-shaped connecting plates thereof are disposed around and connected to an outer wall of the housing body thereof to define the mounting groove thereof.
(15) The arc-shaped connecting plates are disposed at intervals, and a necking portion is disposed at one end of the arc-shaped connecting plates away from a corresponding housing, so that an outer diameter of an opening of each mounting groove is less than the outer diameter of each of the spherical portions. Arc-shaped limiting portions protruding inward are disposed on two opposite sides of an inner wall of each through hole.
(16) In some embodiments, inner threads are disposed on the mounting pieces, outer threads are disposed on the arc-shaped connecting plates, and the inner threads are respectively screwed with the outer threads.
(17) In some embodiments, each housing further comprises support ribs. The support ribs thereof are connected to the housing body thereof and inner walls of the arc-shaped connection plates thereof. Outer surfaces of the support ribs thereof are disposed in arc shapes to form the groove wall of each mounting groove.
(18) In some embodiments, the fixing piece is a stake. The stake defines a mounting hole, and the mounting portion comprises support sheets. A first side of each of the support sheets is connected to first sides of other support sheets. Each two adjacent second sides of the support sheets defines an included angle. The support sheets are in interference fit with an inner wall of the mounting hole.
(19) In some embodiments, the fixing piece is a fixing base, the fixing base defines a fixing surface, and the fixing surface is disposed on the mounting surface. A first rotating buckle is disposed on an outer wall surface of the mounting portion. The fixing portion defines an assembling hole. A second rotating buckle is disposed on a hole wall of the assembling hole. The mounting portion is at least partially inserted into the assembling hole. The first rotating buckle is

engaged with the second rotating buckle by rotating.

(20) In some embodiments, the fixing base comprises a fixing plate. The fixing plate defines the fixing surface and a fixing hole. A fastener is configured to pass through the fixing hole to fix the fixing plate to the mounting surface.

(21) In some embodiments, each of the projection lamps comprises a housing, a lamp board, a projection element, and a projection lens. The housing thereof defines a mounting cavity and a light-transmitting hole. The light-transmitting hole thereof is communicated with the mounting cavity and an outside. The lamp board thereof is mounted in the mounting cavity thereof. The lamp board thereof comprises at least one light source. The projection element thereof is mounted in the mounting cavity thereof and is disposed in a light emitting direction of the at least one light source thereof. The projection lens thereof is mounted in the mounting cavity thereof and is disposed in the light emitting direction of the at least one light source thereof. Light emitted by the at least one light source thereof passes through the projection element thereof, the projection lens thereof and is emitted outward through the light-transmitting hole thereof.

(22) In some embodiments, each of the projection lamps further comprises a spotlight cup mounted on the lamp board thereof and disposed in the light emitting direction of the at least one light source thereof. The projection element thereof is mounted on one side, away from the at least one light source, of the spotlight cup thereof.

(23) In some embodiments, the projection element thereof is a projection film. Each of the projection lamps further comprises a pressing plate. The pressing plate thereof is fixedly connected to the spotlight cup thereof to press the projection element thereof on the spotlight cup thereof.

(24) In some embodiments, the least one light source comprises light sources, and the projection element thereof comprises image structures. The projection lens thereof comprises focusing portions. The light sources are one-to-one corresponding to the image structures. The focusing portions are one-to-one corresponding to the light sources.

(25) Each of the projection lamps further comprises a motor. A main body of the motor thereof is mounted on a rear portion of the lamp board thereof. A driving shaft of the motor thereof passes through the lamp board thereof and is rotatably connected to the projection lens thereof.

(26) The light sources thereof, the image structures thereof, and the focusing portions thereof are circumferentially disposed in a circumferential direction of the driving shaft thereof.

(27) In some embodiments, the connecting piece further comprises a first connecting arm. A middle portion of the first connecting arm is connected to the mounting portion, two opposite ends of the first connecting arm respectively form two connecting portions of the at least two connecting portions, and the first connecting arm is disposed in an arc shape.

(28) In some embodiments, the connecting piece further comprises a second connecting arm. A first end of the second connecting arm is connected to a junction of the first connecting arm and the mounting portion, and a second end of the second connecting arm forms one connecting portion of the at least two connecting portions.

(29) In some embodiments, first reinforcing ribs spaced apart from each other are disposed in at least one of an extending direction of the first connecting arm and an extending direction of the second connecting arm, and/or second reinforcing ribs spaced apart from each other are disposed in at least one of the extending direction of the first connecting arm and the extending direction of the second connecting arm.

(30) In the projection device of the embodiments of the present disclosure, the base comprises the at least two connecting portions, and the at least two connecting portions extend in different directions, so that the at least two projection lamps are allowed to be mounted on one base, and different projection lamps are allowed to project at different positions and angles. Therefore, the projection device of the present disclosure achieves a wider projection coverage and meet various projection needs. Moreover, the at least two projection lamps generate brighter and clearer images, thereby improving projection quality. Further, projections projected by the at least two projection

lamps are allowed to be superimposed to achieve richer colors and layering. In addition, only one base is needs for mounting and fixing of the projection device, and the at least two projection lamps are allowed to be disposed at required positions, which improves mounting efficiency and space utilization.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to clearly describe technical solutions in the embodiments of the present disclosure, the following will briefly introduce the drawings that need to be used in the description of the embodiments or the prior art. Apparently, the drawings in the following description are merely some of the embodiments of the present disclosure, and those skilled in the art are able to obtain other drawings according to the drawings without contributing any inventive labor.

(2) FIG. **1** is a schematic diagram of a projection device according to a first embodiment of the present disclosure.

(3) FIG. **2** is a cross-sectional schematic diagram of the projection device according to the first embodiment of the present disclosure.

(4) FIG. **3** is an exploded schematic diagram of the projection device according to the first embodiment of the present disclosure.

(5) FIG. **4** is a schematic diagram of the projection device according to a second embodiment of the present disclosure.

(6) FIG. **5** is a cross-sectional schematic diagram of the projection device according to the second embodiment of the present disclosure.

(7) FIG. **6** is a schematic diagram of the projection device according to a third embodiment of the present disclosure.

(8) FIG. **7** is a schematic diagram of the projection device according to a fourth embodiment of the present disclosure.

(9) FIG. **8** is a schematic diagram of a connecting piece of the projection device according to the first embodiment of the present disclosure.

(10) FIG. **9** is a schematic diagram of a fixing piece of the present disclosure.

(11) FIG. **10** is a cross-section schematic diagram of a projection lamp of the present disclosure.

(12) FIG. **11** is an exploded schematic diagram of a projection lamp of the present disclosure.

(13) FIG. **12** is a schematic diagram of a mounting piece of the present disclosure.

(14) In the drawings: **1**—projection device; **10**—base; **110**—connecting piece; **111**—main body portion; **1111**—connecting portion; **1112**—mounting portion; **1113**—support sheet; **1114**—first rotating buckle; **1115**—first connecting arm; **1116**—second connecting arm; **1117**—first reinforcing rib; **1118**—second reinforcing rib; **112**—spherical portion; **1121**—hemispherical portion; **1122**—hemispherical cover; **120**—fixing piece; **121**—stake; **121***a*—mounting hole; **122**—fixing base; **122***a*—assembling hole; **1221**—fixing surface; **1222**—fixing plate; **1230**—fixing hole; **1240**—second rotating buckle; **20**—projection lamp; **210**—housing; **210***a*—mounting groove; **210***b*—mounting cavity; **210** *c*—light-transmitting hole; **211**—housing body; **212**—arc-shaped connecting plate; **2121**—necking portion; **213**—supporting rib; **220**—lamp board; **221**—mounting plate **222**—light source; **230**—projection film; **240**—projection lens; **2401**—focusing portion; **250**—spotlight cup; **260**—pressing plate; **270**—motor; **280**—sealing glass plate; **290**—sealing ring; **30**—mounting piece; **30***a*—through hole; **310**—arc-shaped limiting portion.

(15) Realization of purposes, functional features, and advantages of the present disclosure is further explained in conjunction with embodiments and with reference to the accompanying drawings.

DETAILED DESCRIPTION

(16) In order to make the purpose, technical solutions, and advantages of the present disclosure

clear, the following section will further describe the embodiments of the present disclosure in detail with reference to the accompanying drawings.
(17) When the following description refers to the drawings, the same numbers in different drawings refer to the same or similar elements unless otherwise indicated. The implementations described in the following exemplary embodiments do not represent all implementations consistent with the present disclosure. Rather, they are merely examples of apparatus and methods consistent with certain aspects of the present disclosure, as detailed in the appended claims.
(18) It should be understood in the description of the present disclosure that terms such as “first” and “second” are only used for the purpose of description, rather than being understood to indicate or imply relative importance or hint the number of indicated technical features. Thus, the feature limited by “first” and “second” can explicitly or impliedly include at least one feature. Unless otherwise indicated, the term “a plurality of” means two or more. The term “and/or” depict relationship between associated objects and there are three relationships thereon. For example, A and/or B may indicate A exists alone, A and B exist at the same time, and B exists alone. The character “/” generally indicates that the associated object is alternative. The terms “first”, “second”, “third”, etc. in the present disclosure are used only to distinguish similar objects and do not imply a specific ordering of objects.
(19) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art of the present disclosure. The terminology used in the specification is for the purpose of describing specific embodiments only and is not intended to limit the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.
(20) As shown in FIGS. **1-7**, the present disclosure provides a projection device **1**. The projection device **1** of the present disclosure comprises at least two projection lamps **20** and a base **10**. The base **10** is configured to support and fix the at least two projection lamp **20** to ensure that the at least two projection lamps **20** are able to stably project images. The base **10** comprises at least two connecting portions **1111**. The at least two connecting portions **1111** extend in different directions. At least one of the at least two projection lamps **20** is disposed on each of the connecting portions **1111**. Since the at least two projection lamps **20** are respectively mounted on the at least two connecting portions **1111**, and at least two connecting portions **1111** extend in different directions, the at least two projection lamps **20** are allowed to be mounted on one base **10**, and the at least two projection lamps **20** are allowed to project at different positions and angles. Therefore, the projection device **1** of the present disclosure achieves a wider projection coverage and meets various projection needs. Moreover, the at least two projection lamps **20** generate brighter and clearer images, thereby improving projection quality. Further, projections projected by the at least two projection lamps **20** are allowed to be superimposed to achieve richer colors and layering. In addition, only one base **10** is needs for mounting and fixing of the projection device **1**, and the at least two projection lamps **20** are allowed to be disposed at required positions, which improves mounting efficiency and space utilization. Furthermore, the at least two connecting portions **1111** of the present disclosure are detachably connected to at least two projection lamps **20**. Therefore, a user is allowed to mount at least one of the at least two projection lamps **20** on at least one of the at least two connecting portions **1111**.
(21) As shown in FIGS. **2**, **3**, and **5**, the base **10** comprises a connecting piece **110** and a fixing piece **120**. The connecting piece **110** comprises the at least two connecting portions **1111** and a mounting portion **1112**. The at least two connecting portions **1111** are connected to the mounting portion **1112**. The fixing piece **120** is connected to the mounting portion **1112**. The fixing piece **120** fixes the at least two projection lamps **20** to a mounting surface or an external device.
(22) As shown in FIGS. **1-5**, in some embodiment, the at least two connecting portions **1111** are two connecting portions. The connecting piece **110** further comprises a first connecting arm **1115**. A middle portion of the first connecting arm **1115** is connected to the mounting portion **1112**, and two

opposite ends of the first connecting arm **1115** respectively form the two connecting portions **1111**. In this way, stability of ta connection between the two connecting portions **1111** and the mounting portion **1112** is improved. Of course, in other embodiments, each of the connecting portions **1111** is formed by a single connecting arm, and each connecting arm is connected to the mounting portion **1112**, or at least portions of adjacent connecting portions are connected together. Optionally, the first connecting arm **1115** is disposed in an arc shape, which increases a contact area with the mounting portion **1112** and has a higher aesthetic appearance. Of course, in other embodiments, the first connecting arm **1115** may be disposed in a V shape.

(23) As shown in FIGS. **6** and **7**, in some embodiments, the at least two connecting portions are three connecting portions **1111**. The connecting piece **110** further comprises a second connecting arm **1116**. A first end of the second connecting arm **1116** is connected to a junction of the first connecting arm **1115** and the mounting portion **1112**, and a second end of the second connecting arm **1116** forms a third one of the three connecting portions **1111**. In this way, the three connecting portion **1111** are disposed on the connecting piece **110**, so that three projection lamps **20** are allowed to be mounted.

(24) As shown in FIG. **8**, in some embodiments, first reinforcing ribs **1117** spaced apart from each other are disposed in at least one of an extending direction of the first connecting arm **1115** and an extending direction of the second connecting arm **1116**. Further, second reinforcing ribs **1118** spaced apart from each other are disposed in at least one of the extending direction of the first connecting arm **1115** and the extending direction of the second connecting arm **1116**. The first reinforcing ribs **1117** and the second reinforcing ribs **1118** improve the rigidity and stability the first connecting **1115** arm and/or the second connecting arm **1116**, prevent the first connecting **1115** arm and/or the second connecting arm **1116** from bending or deforming when subjected to an external force, and improve the stability of the at least two projection lamps **20**.

(25) It should be noted that the connecting piece **110** and the fixing piece **120** may be an integrally formed component. That is, the connecting piece **110** and the fixing piece **120** may be made of the same material (such as metal, plastic, etc.) through injection molding, casting, or other processes at one time. The integrally formed component has advantages of compact structure, simple manufacturing, and low cost. The connecting piece **110** and the fixing piece **120** may be separated structures, that is, the connecting piece **110** and the fixing piece **120** may be made of the same or different materials, the connecting piece **110** and the fixing piece **120** may be manufactured by the same or different manufacturing processes, and the connecting piece **110** and the fixing piece **120** may be detachably connected together, so that the fixing piece **120** is able to be replaced according to needs.

(26) As shown in FIGS. **1-3** and **6**, in some embodiments, the fixing piece **120** is a stake **121**. A first end of the stake **121** defines a mounting hole **121***a*, and a second end of the stake **121** is inserted into the ground. The stake **121** provides a stable support point, which is convenient for fixing the at least two projection lamps **20** in different scenes, and effectively prevents the at least two projection lamps **20** from shaking or tilting during use. In addition, the at least two projection lamps **20** are able to be fixed at different heights by changing the stake **121** of different lengths, which is convenient for the user to use.

(27) As shown in FIG. **8**, the mounting portion **1112** comprises support sheets **1113**. A first side of each of the support sheets **1113** is connected to first sides of other support sheets **1113** to form a common connecting area. Each two adjacent second sides of the support sheets **1113** defines an included angle and a plurality of included angle are formed. Sizes and directions of the plurality of included angles may be the same or different, but the plurality of included angles together form an outer contour that is approximately circular or annular. The support sheets **1113** are in interference fit with an inner wall of the mounting hole **121***a*. The interference fit connection is simple and firm, and the at least two projection lamps **20** are easily removed and remounted when the at least two projection lamps **20** need to be maintained or replaced.

(28) As shown in FIGS. **4**, **5**, and **7**, in some embodiments, the fixing piece **120** is a fixing base **122**. The fixing base **122** defines a fixing surface **1221**, and the fixing surface **1221** is disposed on the mounting surface (e.g., a desktop, a wall, a floor, etc.). A first rotating buckle **1114** is disposed on an outer wall surface of the mounting portion **1112**. The fixing portion defines an assembling hole **122***a*. A second rotating buckle **1240** is disposed on a hole wall of the assembling hole **122***a*. The mounting portion **1112** is at least partially inserted into the assembling hole **122***a*. The first rotating buckle **1114** is engaged with the second rotating buckle **1240** by rotating. When assembly, the user only needs to insert the mounting portion **1112** into the assembling hole **122***a* and rotate the mounting portion **1112** to an appropriate position to achieve a secure connection. By such arrangement, no additional tool or fastening piece is required, thereby greatly improving mounting efficiency. Similar to the assembling process, during disassembly, the mounting portion **1112** and the fixing member **120** are easily separated by rotating, which facilitates maintenance and replacement of the projection device, thereby reducing maintenance costs and time.
(29) As shown in FIG. **9**, furthermore, the fixing base **122** comprises a fixing plate **1222**. The fixing plate **1222** defines the fixing surface **1221** and a fixing hole **1230**. A fastener (e.g., a bolt, a screw, etc.) is configured to pass through the fixing hole to fix the fixing plate **1222** to the mounting surface, which ensures a firm connection between the fixing base **122** and the mounting surface. A connection method thereof has high tensile strength and shear strength, is able to withstand large external forces and vibrations, and ensures the stability and safety of the at least two projection lamps **20**.
(30) As shown in FIGS. **10-11**, each of the projection lamps **20** comprises a housing **210**, a lamp board **220**, a projection element and a projection lens **240**. The housing **210** thereof defines a mounting cavity **210***b* and a light-transmitting hole **210***c*. The light-transmitting hole **210***c* thereof is communicated with the mounting cavity **210***b* and an outside. The lamp board **220** thereof is mounted in the mounting cavity **210***b* thereof. The lamp board **220** thereof comprises at least one light source **222**. The projection element thereof is mounted in the mounting cavity **210***b* thereof and is disposed in a light emitting direction of the at least one light source **222** thereof. The projection lens **240** thereof is mounted in the mounting cavity **210***b* thereof and is disposed in the light emitting direction of the at least one light source **222** thereof. Light emitted by the at least one light source **222** thereof passes through the projection element thereof, the projection lens **240** thereof and is emitted outward through the light-transmitting hole **210***c* thereof.
(31) As shown in FIGS. **10-11**, for ease of illustration, the present disclosure takes one of the least two projection lamps **20** as an example for further description. The housing **210** serves as an outer shell of the projection lamp **20**, protecting internal components of the projection lamp **20** from interference from an external environment. Furthermore, protrusions are disposed on an outer surface of the housing **210** to increase a contact area between the housing **210** and air, thereby improving a heat dissipation effect. The housing **210** comprises a rear housing and a front housing. The rear housing defines an accommodating groove with an opening on one side, and the front housing is screwed with the rear housing and covers the opening of the accommodating groove. The front housing defines the light-transmitting hole **210***c*. A sealing glass plate **280** is mounted at the light-transmitting hole **210***c* to seal the mounting cavity **210***b* to prevent dust or water vapor from entering the mounting cavity **210***b*. Further, sealing rings **290** are disposed on two opposite sides of the sealing glass plate **280**. That is, a first sealing ring **290** is disposed between the sealing glass plate **280** and the front housing, and a second sealing ring **290** is disposed between the sealing glass plate **280** and the rear housing after the front housing is fixed with the rear housing. In some embodiments, the housing **210** may comprise a rear cover, a middle housing, and a front cover. The middle housing is in a cylindrical structure with openings at two ends thereof. The rear cover and the front cover are respectively mounted at the two ends of the middle housing and cover the two openings of the middle housing. The front cover defines the light-transmitting hole **210***c*.
(32) As shown in FIG. **11**, the light panel **220** comprises a mounting plate **221** and light sources

222 mounted on the mounting plate **221**. The light sources **222** may be LED lamp beads. The projection element may carry a specific image, image structure or text. The projection element may be a light sheet, a film, a liquid crystal light valve, etc. Specifically, the light emitted by the light sources **222** is converted into a light image after passing through the projection element, and the light image is directed to the projection lens **240**. The projection lens **240** is configured to amplify the light image and project an amplified light image out (such as projecting the amplified light image onto a projection surface such as the ground, the wall, a curtain, etc. to form an projected image).
(33) Furthermore, the projection lens **240** may comprise only a convex lens. In other embodiments, the projection lens **240** may include lenses. For example, the projection lens **240** comprises a first lens, a second lens, and a third lens coaxially disposed in sequence along the light emitting direction. The first lens and the third lens have positive optical power, and the second lens has negative optical power, which are not limited thereto. The structure and principle of the projection lens **240** have been disclosed in the relevant art, and are not depicted in details in the present disclosure.
(34) As shown in FIGS. **10-11**, in some embodiments, the projection lamp **20** further comprises a spotlight cup **250** mounted on the lamp board **220** and disposed in the light emitting direction of the light sources **222** The projection element is mounted on one side, away from the light sources **222**, of the spotlight cup **250**. The spotlight cup **250** and the mounting plate **221** are connected by screws. The spotlight cup **250** is also called as a reflective cup, a focusing reflector, a light cup, etc. The spotlight cup **250** is configured to reflect and focus the light, and has a an entrance and exit opposite to the entrance. A size of the entrance is less than a size of the exit, so that a focusing cavity defined inside the spotlight cup **250** is of a trapezoidal structure. In addition, a reflective layer is disposed on an inner wall surface of the spotlight cup **250** to reflect light to reflect and focus incident light, so that outgoing light is roughly uniformly emitted in parallel. In this way, a divergence range of the emitted light is reduced, forming a small-angle light with good brightness and uniformity, so that the emitted light passes through the projection element and the projection lens **240** as much as possible, forming a better projection image.
(35) It should be noted that the reflective layer may be a reflective coating evenly coated on the inner wall surface of the spotlight cup **250**. The reflective layer may also be a reflective film attached to the inner wall surface of the spotlight cup **250**. Alternatively, a reflective material (such as aluminum, silver, silicon dioxide, etc.) is deposited on the inner wall surface of the spotlight cup **250** through evaporation, magnetron sputtering, physical vapor deposition, chemical vapor deposition, or other coating techniques to form the reflective layer on the inner wall surface of the spotlight cup **250**.
(36) As shown in FIGS. **10-11**, in some embodiments, the projection element is a projection film **230**. The projection lamp **20** further comprises a pressing plate **260**. The pressing plate **260** is fixedly connected to the spotlight cup **250** to press the projection film **230** on the spotlight cup **250**. The pressing plate **260** is fixedly connected to the spotlight cup **250**, and the pressing plate **260** firmly presses the projection film **230** onto the spotlight cup **250**, preventing the projection film **230** from moving or loosening during a projection process, ensuring clarity and stability of the projected image, and avoiding image distortion or blurring caused by position changes of the projection film **230**. The pressing plate **260** is fixedly connected to the spotlight cup **250**, which further ensures that a contact between the projection film **230** and the spotlight cup **250** is tight and uniform, thus reducing leakage or scattering of light between the projection film **230** and the spotlight cup **250**, improving the utilization rate of light, and improving the projection effect.
(37) It is understood that the projection film **230** comprises image structures. The pressing plates defies restrict holes corresponding to the image structures. The pressing plate **260** and the spotlight cup **250** press an edge of the projection film **230**, which ensures that the image structures remains flat during the projection process and avoid image distortion or blurring caused by curling or

deformation of the projection film **230**. In addition, the restrict holes prevent the light from being reflected or scattered inside the pressing plate **260**, so light utilization efficiency is not reduced, and darkening of the projected image or an appearance of light spots are avoided. The pressing plate **260** is made of polymethyl methacrylate (PMMA), polystyrene (PS), polycarbonate (PC) or glass, which have high light transmittance, good heat resistance and impact resistance.
(38) As shown in FIG. **11**, in some embodiments, the light sources **222 222** are mounted on the mounting plate **221**, the projection film **230** comprises the image structures, and the projection lens **240** comprises focusing portions **2401**. The light sources **222** are one-to-one corresponding to the image structures in the light emitting direction. The focusing portions **2401** are one-to-one corresponding to the light sources **222**. It should be noted that the focusing portions **2401** are raised structures disposed on the projection lens **240**. The projection lamp **20** further comprises a motor **270** A main body of the motor **270** is mounted on a rear portion of the mounting plate **221**. A driving shaft of the motor **270** passes through the mounting plate **221** and is rotatably connected to the projection lens **240**. The light sources **222**, the image structures, and the focusing portions **2401** are circumferentially disposed in a circumferential direction of the driving shaft. The motor **270** drives the projection lens **240** to rotate, so that light image that is originally static presents a dynamic effect, increasing interest and ornamental value of the projected image. Furthermore, parameters such as a rotating speed, a direction, and a rotating time of the motor **270** are controlled by programming, so that the projection lamp **20** is able to perform dynamic projection according to a preset animation effect to meet diverse application needs.
(39) Furthermore, the mounting plate **221** of the lamp board **220** is a circuit board, and the projection lamp **20** further comprises a conductive wire, a first end of the conductive wire is electrically connected to the circuit board and a second end of the conductive wire is connected to a plug, so that the light sources **222** and the motor **270** are powered by an external power supply.
(40) Remaining structures of the projection device **1** are introduced below.
(41) In some embodiments, the at least two projection lamps **20** are respectively rotatably connected to the at least two connecting portions **1111**, which allows the at least two projection lamps **20** to rotate freely in different directions, so that projection angles and positions thereof are easily adjusted.
(42) In some embodiments, the at least two connecting portions **1111** comprise connection positions respectively connected to the at least two projection lamps **20**. In a front-rear direction of the projection device **1**, the connection positions are located on a same plane. Since the at least two projection lamps **20** are located on the same plane, projection lights of the at least two projection lamps **20** are not blocked by each other, ensuring that each of the projection lamps **20** is able to independently and completely display the projected image or a light effect thereof. In addition, since light blocking is avoided, there is no need to frequently adjust a position or an angle of each of the projection lamps **20** during a debugging process, thereby saving debugging time.
(43) As shown in FIGS. **2**, **3**, and **8**, in some embodiments, the connecting piece **110** further comprises spherical portions **112**. Each of the spherical portions **112** are connected to one end of a corresponding one of the at least two connecting portions **1111**. Each housing **210** define a mounting groove **210***a*, and a groove wall of each mounting groove **210***a* is spherical. The projection device **1** of the present disclosure further includes mounting pieces **30**. Each of the mounting pieces **30** defines a through hole **30***a*. An aperture of each through hole **30***a* is less than an outer diameter of each of the spherical portions **112**. Each of the mounting pieces **30** is detachably connected to each housing **210** to rotatably connect each of the spherical portions **112** in each mounting groove **210***a*. Each of the connecting portions **1111** extends outward from each through hole **30***a*.
(44) Each of the spherical portions **112** is connected to the one end of the corresponding one of the at least two connecting portions **1111** and is rotatably fixed in the mounting groove **210***a* of each housing **210**, so that each of the connecting portions **1111** is rotatable freely within a certain range,

and the projection angle of each of the projection lamps **20** is easily adjusted, realizing multi-angle projection and meeting needs of different scenes. In addition, the groove wall of each mounting groove **210***a* is spherically disposed, which is tightly matched with each of the spherical portions **112**, ensuring the stability of the connecting piece **110** during a rotating process. Each of the mounting pieces **30** defines the through hole **30***a*, and the aperture of each through hole **30***a* is less than the outer diameter of each of the spherical portions **112**, which effectively prevents the spherical portions **112** from falling off during the assembly process and improves the reliability of a connection thereof.

(45) As shown in FIG. **8**, in some embodiments, the connecting piece **110** comprises a main body portion **111** and hemispherical coves **1122**. The main body portion **111** comprises the mounting portion **1112**, the at least two connecting portions **1111**, and hemispherical portions **1121** respectively connected to the at least two connecting portions **1111**. The hemispherical covers **1122** are respectively detachably connected to the hemispherical portions **1121** to form the spherical portions **112**. It is understood that, since the aperture of the through hole **30***a* of each of the mounting pieces **30** is less than the outer diameter of each of the spherical portions **112**, a complete spherical portion **112** is unable to directly pass through each through hole **30***a*. At this time, each of the hemispherical portions **1121** (as a part of each of the spherical portions **112**) passes through each through hole **30***a* first. After each of the hemispherical portions **1121** passes through each through hole **30***a*, each of the hemispherical covers **1122** is relatively fixed to each hemispherical portion **1121** to form each spherical portion **112**. Such a step-by-step mounting method makes the assembly process simpler and solves a problem that each of the spherical portions **112** is unable to pass through each through hole **30***a* because the spherical portions **112** are too large as a whole. Of course, in other embodiments, each of the mounting pieces **30** may be separate structures. For instance, each of the mounting pieces **30** is designed as two parts, and the two parts are enclosed and fixed as a whole on two sides of each of the connecting portions **1111**, which is not limited thereto.

(46) Optionally, the hemispherical covers **1122** are respectively connected to the hemispherical portions **1121** through screws, a connection method thereof is simple and has high stability. Of course, in other embodiments, a snap-on connection method may be adopted to connected the hemispherical covers **1122** and the hemispherical portions **1121**.

(47) Alternatively, the housing **210** of each of the projection lamp **20** is connected to each of the spherical portions **112**, and each mounting groove **210** *a* is formed at the one end of the corresponding one of the at least two connecting portion **1111** The present disclosure does not limit the specific connection method of the at least two projection lamps **20** and each housing **210** thereof.

(48) As shown in FIGS. **10-11**, each housing **210** comprises a housing body **211** and arc-shaped connecting plates **212**. The arc-shaped connecting plates **212** thereof are disposed around and connected to an outer wall of the housing body **211** thereof to define the mounting groove **210***a* thereof. The arc-shaped connecting plates **212** of each housing **210** are disposed at intervals so that the arc-shaped connecting plates **212** undergo a certain degree of elastic deformation when subjected to the external force. When each of the spherical portion **112** enters each mounting groove **210***a*, corresponding arc-shaped connecting plates **212** expand slightly outward to match with an outer diameter of each of the spherical portions **112**, thereby ensuring that each of the spherical portions **112** is allowed to be smoothly embedded in each mounting groove **210***a*. A necking portion **2121** is disposed at one end of the arc-shaped connecting plates **212** away from a corresponding housing **210**. Each necking portion **2121** plays a role of a “clip”. Once each of the spherical portions **112** completely enters each mounting groove **210***a*, each necking portion **2121** tightly hold each of the spherical portions **112** to prevent it from accidentally detaching from each mounting groove **210***a*. Optionally, the mounting pieces **30** comprise internal threads, the arc-shaped connecting plate **212** of each housing comprises outer threads, and the internal threads are

respectively screwed with the outer threads. On one hand, the threads make the projection device **1** easy to disassemble, and on the other hand, each of the mounting pieces **30** provides a support force to the corresponding arc-shaped connecting plates on outer sides of the corresponding arc-shaped connecting plates **212**, thereby improving the stability of the arc-shaped connecting plates **212** in fixing the spherical portions **112**.
(49) As shown in FIG. **12**, furthermore, arc-shaped limiting portions **310** protruding inward are disposed on two opposite sides of an inner wall of each through hole **30***a*. The arc-shaped limiting portions **310** of each of the mounting pieces **30** limit a rotation range of each of the spherical portions **112** in each mounting groove **210***a*, preventing each of the spherical portions **112** from excessive rotation or deviation from a predetermined position, ensuring stability and accuracy of each of the projection lamps **20** after mounting, and avoiding projection deviation or equipment damage caused by improper rotation.
(50) In some embodiments, each housing **210** further comprises support ribs **213**. The support ribs **213** thereof are connected to the housing body **211** thereof and inner walls of the arc-shaped connection plates thereof. Outer surfaces of the support ribs **213** thereof are disposed in arc shapes to form the groove wall of each mounting groove **210***a*. The support ribs **213** thereof are connected to an inner wall of each housing body **211** and inner wall of corresponding arc-shaped connecting plates **212**, which improve the stability of the connection between each housing body **211** and the corresponding arc-shaped connecting plate **212**. The support ribs **213** of each housing **200** form the mounting groove **210***a* of a spherical shape. Further, the support ribs **213** of each housing **210** provide uniform support force, reduce deformation and damage risks of each housing **210** when subjected to the external force.
(51) In the drawings of the embodiments, the same or similar numbers correspond to the same or similar components. In the description of the present disclosure, it should be understood that terms such as “upper”, “lower”, “left”, “right” etc. indicate direction or position relationships shown base **10***d* on the drawings, and are only intended to facilitate the description of the present disclosure and the simplification of the description rather than to indicate or imply that the indicated device or element must have a specific direction or constructed and operated in a specific direction. Therefore, the terms used to describe positional relationships in the drawings are only for illustrative purposes and cannot be construed as limitations of the present disclosure. For those of ordinary skill in the art, the specific meanings of the above terms can be understood according to specific circumstances.
(52) The above are only optional embodiments of the present disclosure and are not intended to limit the present disclosure. Any modifications, equivalent substitutions, and improvements made within the spirit and principles of the present disclosure shall be included in the protection scope of the present disclosure.

Claims

1. A projection device, comprising: at least two projection lamps; and a base; wherein the base comprises at least two connecting portions, the at least two connecting portions extend in different directions, and at least one of the at least two projection lamps is disposed on each of the connecting portions; wherein a connecting piece comprises the at least two connecting portions and a mounting portion, the at least two connecting portions are connected to the mounting portion, the connecting piece further comprises spherical portions and mounting pieces, each of the spherical portions are connected to one end of a corresponding one of the at least two connecting portions; wherein each of the projection lamps comprises a housing, each housing define a mounting groove, and a groove wall of each mounting groove is spherical; wherein each of the mounting pieces defines a through hole, an aperture of each through hole is less than an outer diameter of each of the spherical portions, each of the mounting pieces is detachably connected to each housing to

rotatably connect each of the spherical portions in each mounting groove, and each of the connecting portions extends outward from each through hole; and wherein the connecting piece comprises a main body portion and hemispherical coves; wherein the main body portion comprises the mounting portion, the at least two connecting portions, and hemispherical portions respectively connected to the at least two connecting portions; wherein the hemispherical covers are respectively detachably connected to the hemispherical portions to form the spherical portions.
2. The projection device according to claim 1, wherein the base further comprises a fixing piece, the fixing piece is connected to one end of the mounting portion away from the at least two projection lamps, and the fixing piece fixes the at least two projection lamps to a mounting surface or an external device.
3. The projection device according to claim 2, wherein the at least two projection lamps are respectively rotatably connected to the at least two connecting portions.
4. The projection device according to claim 3, wherein the at least two connecting portions comprise connection positions respectively connected to the at least two projection lamps; wherein in a front-rear direction of the projection device, the connection positions are located on a same plane.
5. The projection device according to claim 1, wherein the hemispherical covers are respectively connected to the hemispherical portions through screws.
6. The projection device according to claim 1, wherein each housing comprises a housing body and arc-shaped connecting plates, and the arc-shaped connecting plates thereof are disposed around and connected to an outer wall of the housing body thereof to define the mounting groove thereof; wherein the arc-shaped connecting plates are disposed at intervals, and a necking portion is disposed at one end of the arc-shaped connecting plates away from a corresponding housing, so that an outer diameter of an opening of each mounting groove is less than the outer diameter of each of the spherical portions, arc-shaped limiting portions protruding inward are disposed on two opposite sides of an inner wall of each through hole.
7. The projection device according to claim 6, wherein inner threads are disposed on the mounting pieces, outer threads are disposed on the arc-shaped connecting plates, and the inner threads are respectively screwed with the outer threads.
8. The projection device according to claim 6, wherein each housing further comprises support ribs, the support ribs thereof are connected to the housing body thereof and inner walls of the arc-shaped connection plates thereof, and outer surfaces of the support ribs thereof are disposed in arc shapes to form the groove wall of each mounting groove.
9. The projection device according to claim 2, wherein the fixing piece is a stake, the stake defines a mounting hole, the mounting portion comprises support sheets, first sides of each of the support sheets are connected to each other, each two adjacent second sides of the support sheets defines an included angle, and the support sheets are in interference fit with an inner wall of the mounting hole.
10. The projection device according to claim 2, wherein the fixing piece is a fixing base, the fixing base defines a fixing surface, and the fixing surface is disposed on the mounting surface, a first rotating buckle is disposed on an outer wall surface of the mounting portion, the fixing portion defines an assembling hole, a second rotating buckle is disposed on a hole wall of the assembling hole, the mounting portion is at least partially inserted into the assembling hole, and the first rotating buckle is engaged with the second rotating buckle by rotating.
11. The projection device according to claim 10, wherein the fixing base comprises a fixing plate, the fixing plate defines the fixing surface and a fixing hole, and the fixing hole is configured for a fastener to pass through, so that the fixing plate is fixed to the mounting surface.
12. The projection device according to claim 2, wherein each of the projection lamps comprises a housing, a lamp board, a projection element, and a projection lens; wherein the housing thereof defines a mounting cavity and a light-transmitting hole, the light-transmitting hole thereof is

communicated with the mounting cavity and an outside, the lamp board thereof is mounted in the mounting cavity thereof, the lamp board thereof comprises at least one light source; the projection element thereof is mounted in the mounting cavity thereof and is disposed in a light emitting direction of the at least one light source thereof; the projection lens thereof is mounted in the mounting cavity thereof and is disposed in the light emitting direction of the at least one light source thereof, and light emitted by the at least one light source thereof passes through the projection element thereof, the projection lens thereof and is emitted outward through the light-transmitting hole thereof.
13. The projection device according to claim 12, wherein each of the projection lamps further comprises a spotlight cup mounted on the lamp board thereof and disposed in the light emitting direction of the at least one light source thereof, and the projection element thereof is mounted on one side, away from the at least one light source, of the spotlight cup thereof.
14. The projection device according to claim 13, wherein the projection element thereof is a projection film, each of the projection lamps further comprises a pressing plate, and the pressing plate thereof is fixedly connected to the spotlight cup thereof to press the projection element thereof on the spotlight cup thereof.
15. The projection device according to claim 12, wherein the least one light source comprises light sources, the projection element thereof comprises image structures, the projection lens thereof comprises focusing portions, the light sources are one-to-one corresponding to the image structures, and the focusing portions are one-to-one corresponding to the light sources, wherein each of the projection lamps further comprises a motor, a main body of the motor thereof is mounted on a rear portion of the lamp board thereof, and a driving shaft of the motor thereof passes through the lamp board thereof and is rotatably connected to the projection lens thereof; wherein the light sources thereof, the image structures thereof, and the focusing portions thereof are circumferentially disposed in a circumferential direction of the driving shaft thereof.
16. The projection device according to claim 2, wherein the connecting piece further comprises a first connecting arm, a middle portion of the first connecting arm is connected to the mounting portion, and two opposite ends of the first connecting arm respectively form two connecting portions of the at least two connecting portions, and the first connecting arm is disposed in an arc shape.
17. The projection device according to claim 16, wherein the connecting piece further comprises a second connecting arm, a first end of the second connecting arm is connected to a junction of the first connecting arm and the mounting portion, and a second end of the second connecting arm forms one connecting portion of the at least two connecting portions.
18. The projection device according to claim 17, wherein first reinforcing ribs spaced apart from each other are disposed in at least one of an extending direction of the first connecting arm and an extending direction of the second connecting arm; and/or second reinforcing ribs spaced apart from each other are disposed in at least one of the extending direction of the first connecting arm and the extending direction of the second connecting arm.